

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Cummins® electronic service tool, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

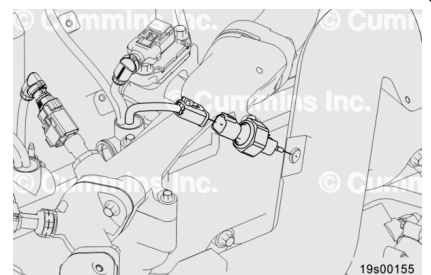
- Disconnect the batteries. See equipment manufacturer service information.

Remove

The intake manifold absolute pressure/temperature sensor measures the pressure/temperature of the intake manifold.

Lift up on the locking tab and pull the electrical connectors apart.

Remove the sensor from the engine by turning the sensor **counterclockwise**.



Clean and Inspect for Reuse

⚠ WARNING ⚠

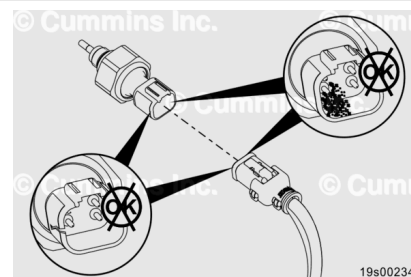
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

If the intake manifold absolute pressure and temperature sensor has frozen, defrost the sensor using warm air that is below 60°C [140°F].

Dry with compressed air.

Inspect the engine wiring harness connector and the sensor for the following:

- Cracked or broken connector shell
- Missing or damaged connector seals
- Dirt, debris, or moisture in or on the connector pins
- Corroded, bent, broken, pushed back, or expanded pins.



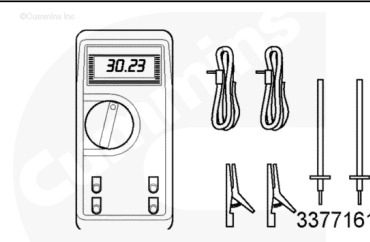
Repair or replace the engine wiring harness connector and sensor.

Test

Disconnect the intake manifold absolute pressure/temperature sensor from the engine wiring harness.

Measure the resistance between the temperature signal pin and return pin at the intake manifold absolute pressure/temperature sensor connector.

If the resistance does **not** meet the specification, replace the intake manifold absolute pressure/temperature sensor.



Temperature vs Resistance

Temperature		Resistance (Ohms)	
°C	°F	MIN	MAX
-50	-58	607,265	843,786
-40	-40	302,255	419,979
-30	-22	157,596	218,978
-20	-4	85,643	119,000

Temperature vs Resistance			
Temperature		Resistance (Ohms)	
-10	14	48,298	67,109
0	32	30,683	36,605
10	50	18,472	22,036
20	68	11,462	13,674
30	86	7456	8578
40	104	4876	5610
50	122	3263	3754
60	140	2233	2569
70	158	1560	1795
80	176	1111	1278
90	194	814	917
100	212	605	669
110	230	450	502
120	248	339	383
130	266	259	396
140	284	201	231
150	302	157	183

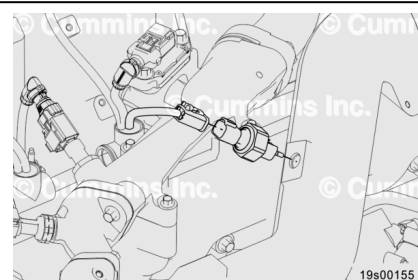
Install

Make sure the new sensor has an o-ring installed on it.

Install the new sensor into the engine by turning the sensor **clockwise**.

Torque Value: 24 n•m [212 in-lb]

Push the connectors together until they lock.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.
- Use the recommended Cummins® electronic service tool, or equivalent, to reset the fuel tables if intake manifold absolute pressure/temperature sensor is replaced. Refer to Procedure 005-263 in Section 5. (</qs3/pubsys2/xml/en/procedures/1016/1016-005-263.html>)
- Operate the engine. Check for proper operation.